



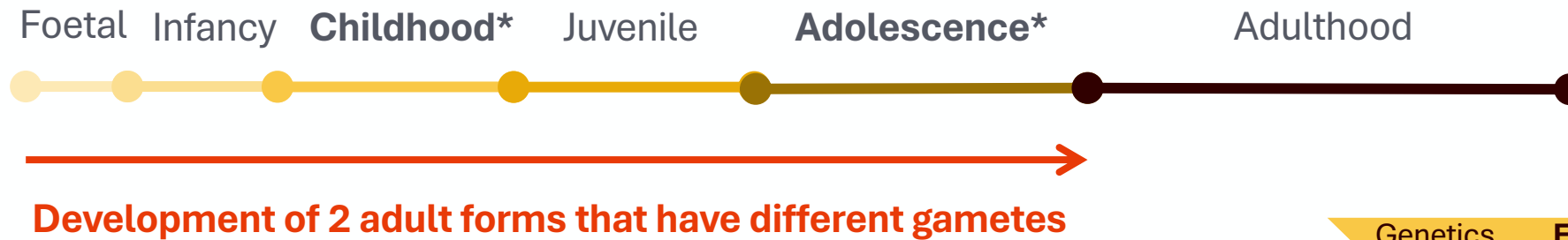
The Kiss (Klimt, 1908)

Reproduction II

Dr. Laura van Holstein

lavanho@emory.edu

ANT215

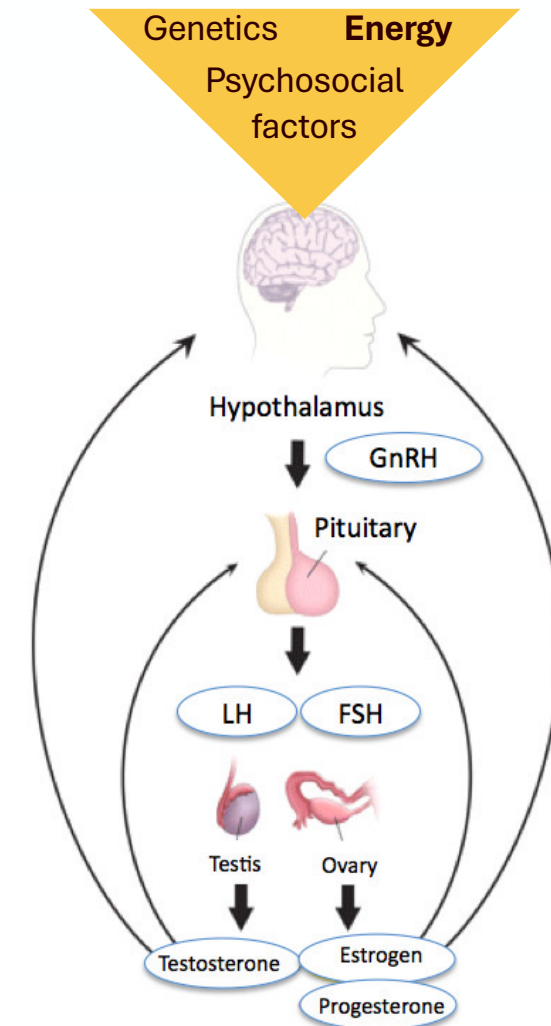


What is sex?

1. Chromosomal, gonadal, endocrine, genital, and phenotypic sex
2. There are strong correlations between these, but $R^2 \neq 1$!
3. There is no absolute binary at any of these levels apart from
 - the existence of the sex chromosomes (ish)
 - Gametes (no spergs or speggs) – but some people could *produce* both

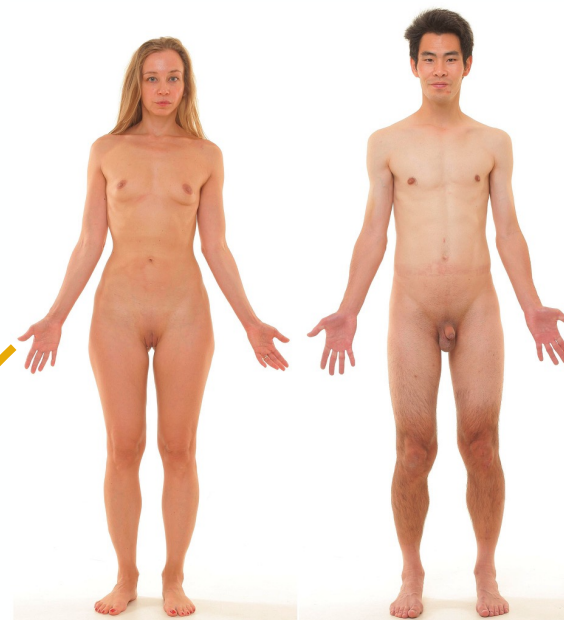
How does sex develop?

1. Chromosomal sex is set at fertilisation. The SRY gene on the Y-chromosome plays a key role in driving male differentiation
2. Followed by 6-week *indifferent period*. Some structures are bipotential (gonads and external genitalia); others aren't and degrade depending on SRY gene activation (Müllerian & Wolffian ducts)
3. 6+ weeks: sexual differentiation, driven largely by SRY – testes – T in males.
4. Minipuberty in infancy
5. Adolescence: divergent somatic & sexual growth patterns





Development of 2 adult forms that have different gametes

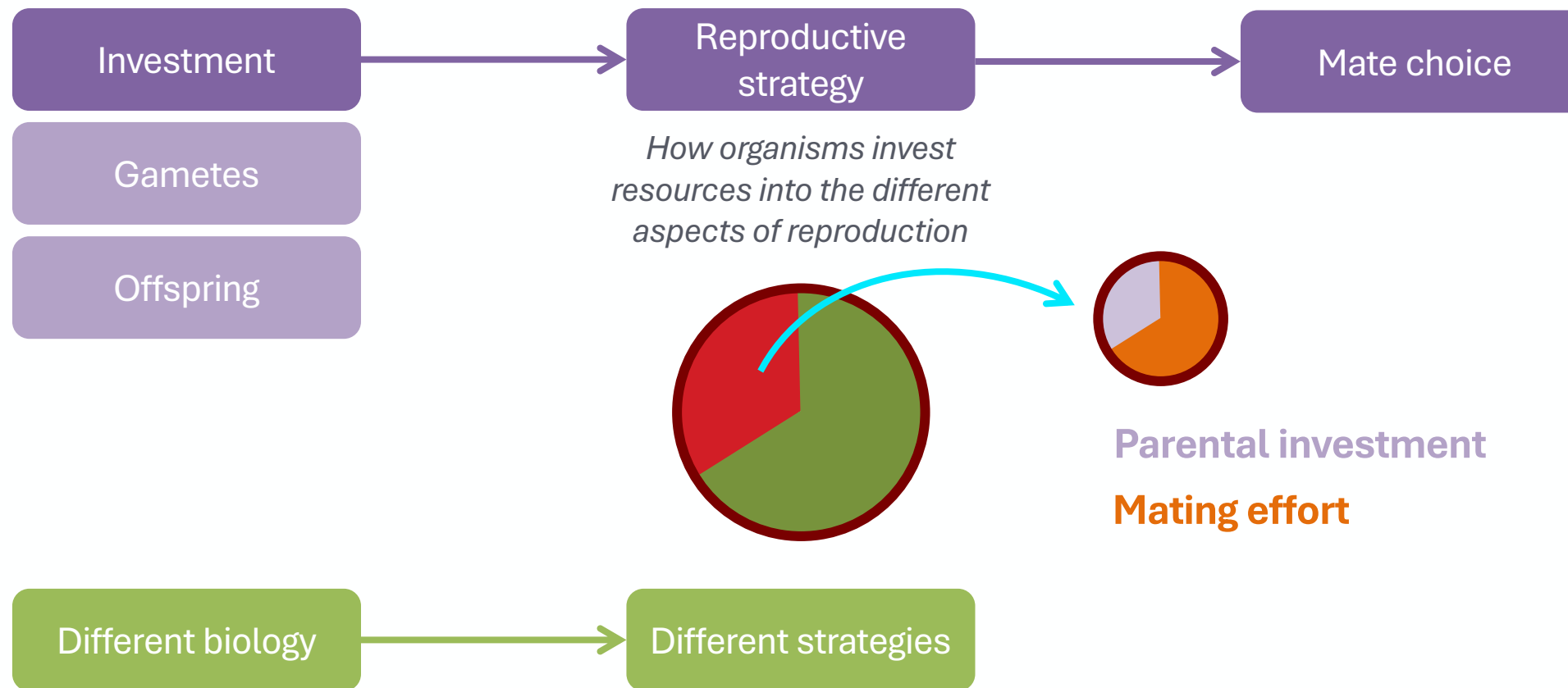


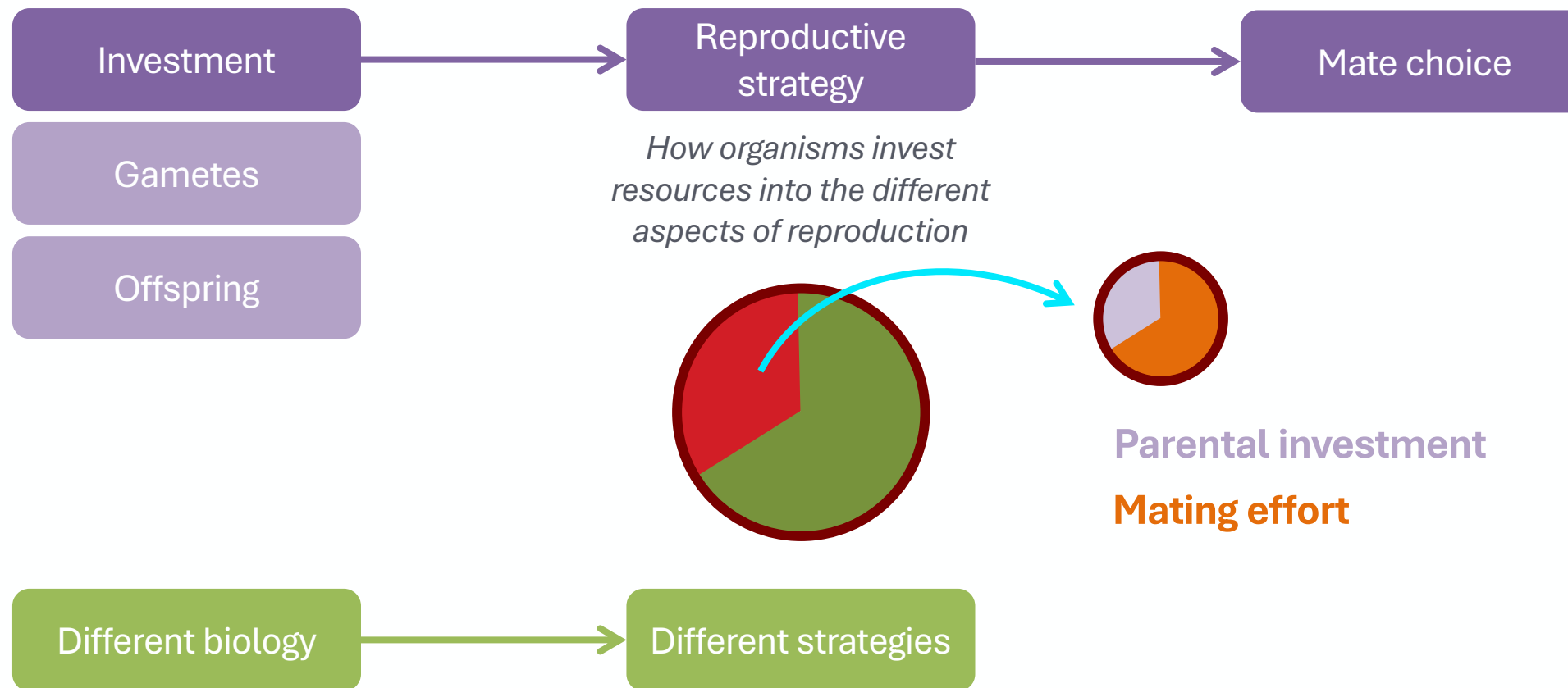
Different biology

Different strategies

Finally, the point of life (from an evolutionary POV):
reproduction

Goal: maximise fitness
In practice: maximise
number of offspring (who bear
offspring who bear offspring who bear... ?)



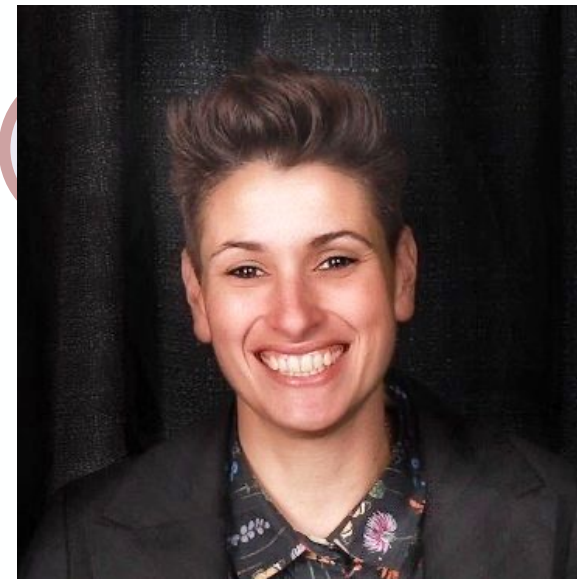
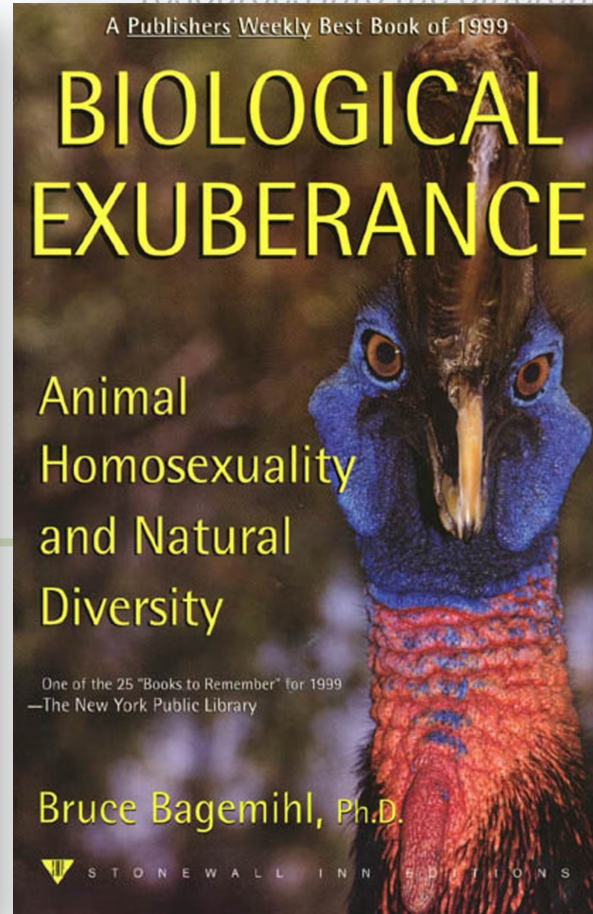
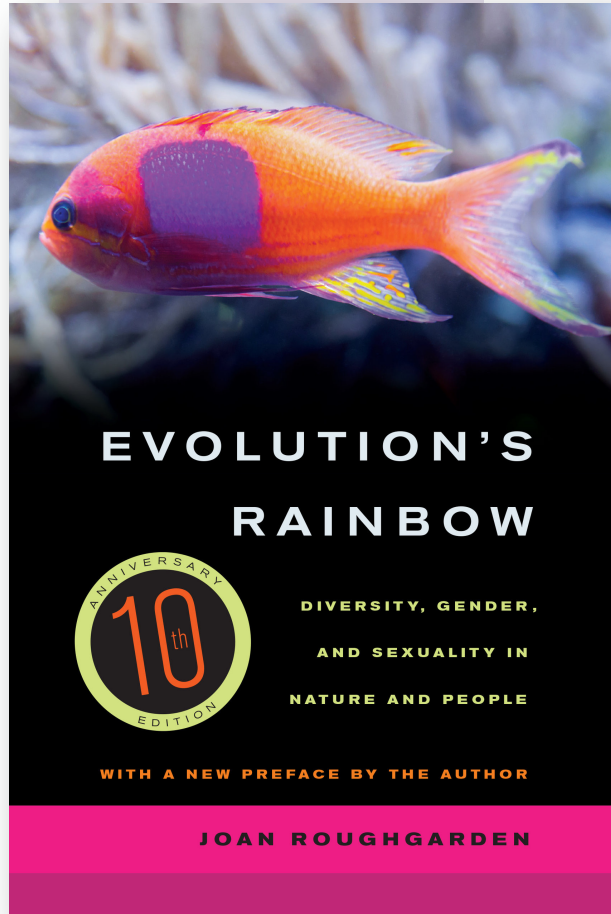


Investment

Reproductive
strategy

Mate choice

Gametes

*How organisms invest
resources into the different*

Prof. Benítez: *Nature, Nurture, Sexuality* (may be offered in 2026-27; keep an eye out on Course Atlas)

Investment

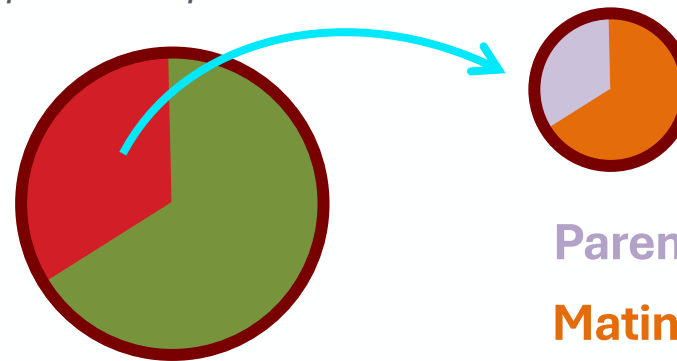
Gametes

Offspring

Reproductive strategy

How organisms invest resources into the different aspects of reproduction

Mate choice



Parental investment
Mating effort

Different biology

Different strategies



All sexually reproducing organisms



Mammals



Humans

Do humans behave as expected for a mammal in their ecosystems?

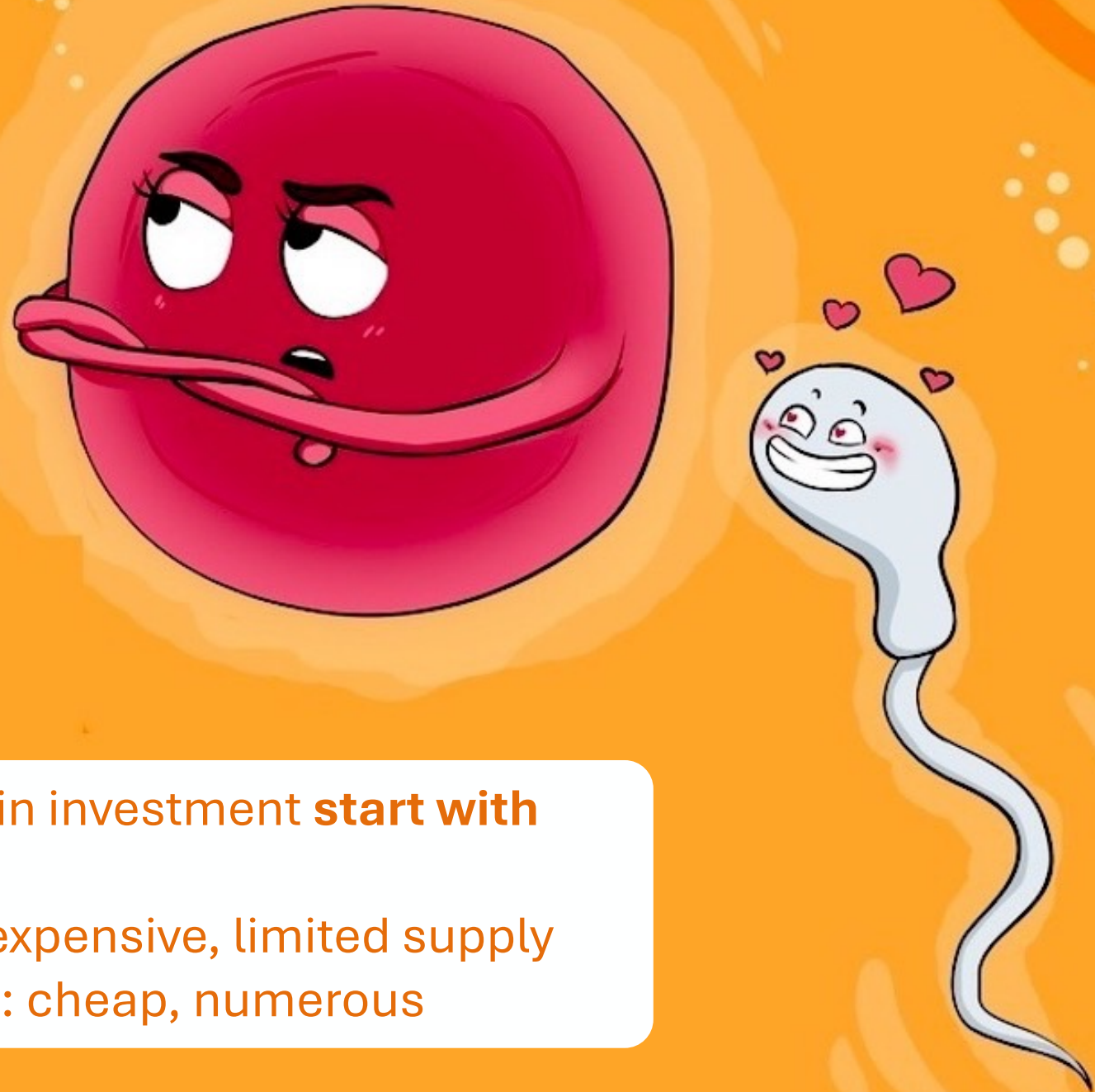


Anisogamy

Differences in the size and form of gametes

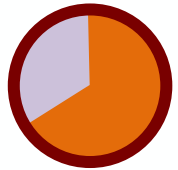
Oogamy

Large non-motile eggs; small, motile sperm



Differences in investment **start with gametes**

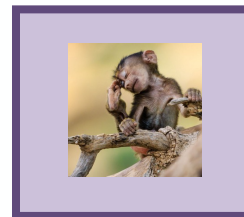
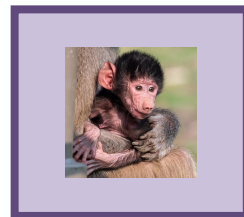
- Eggs: expensive, limited supply
- Sperm: cheap, numerous



Parental investment vs Mating effort

- **Parental investment** (Trivers 1972):

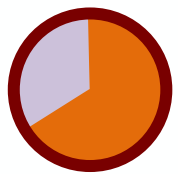
lection. I first define parental investment as *any investment by the parent in an individual offspring that increases the offspring's chance of surviving (and hence reproductive success) at the cost of the parent's ability to invest in other offspring*. So defined, parental investment includes the meta-



Large egg
Internal gestation
Lactation
Provisioning



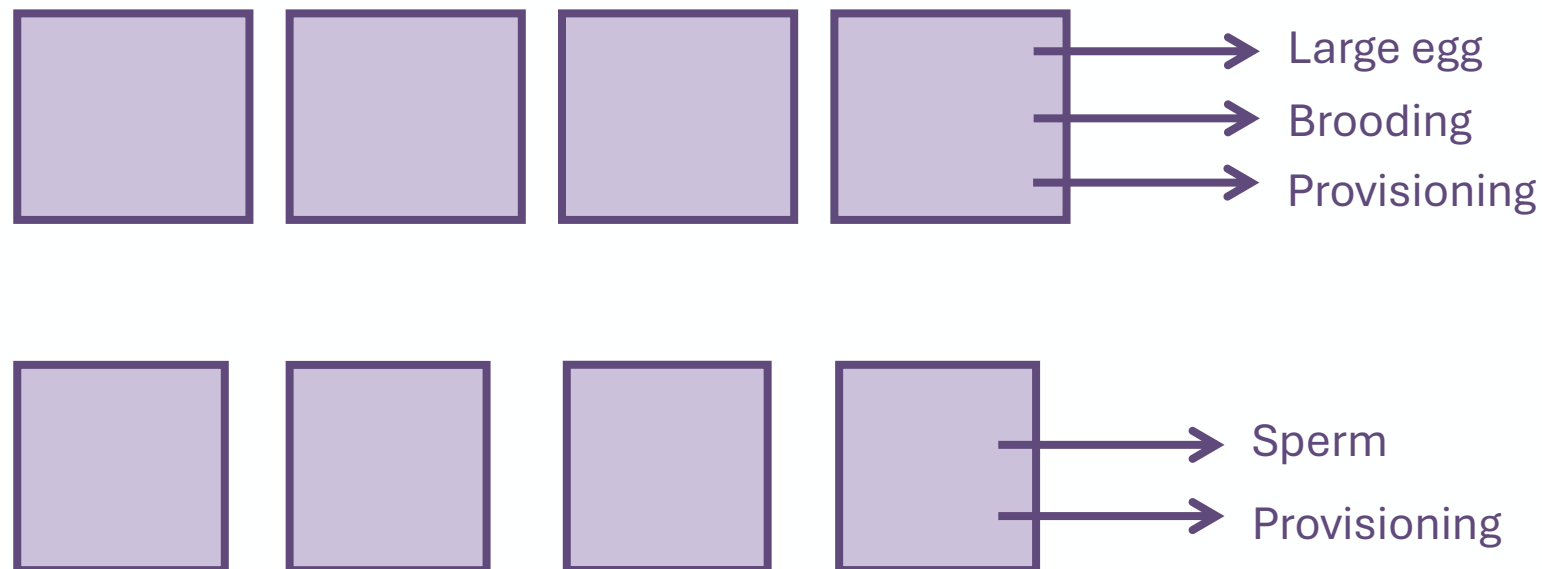
Sperm

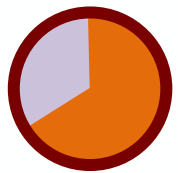


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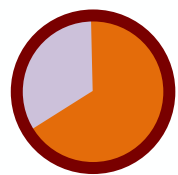


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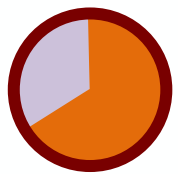
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Maynard Smith (1977): ESS model of parental investment
Costs of trade-offs for both sexes lead to 4 strategies:

1. **Both desert**
2. **Female care** (if male can increase reproductive success by mating again, and one parent is much better than none)
3. **Male care** (if female can lay more eggs if she does not invest in caring, and one parent is much better than none)
4. **Biparental care** (if two parents can raise > twice as many young as one or if chance that deserting parent remates is small)



Parental investment vs Mating effort

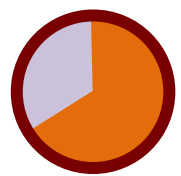
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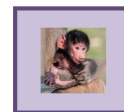
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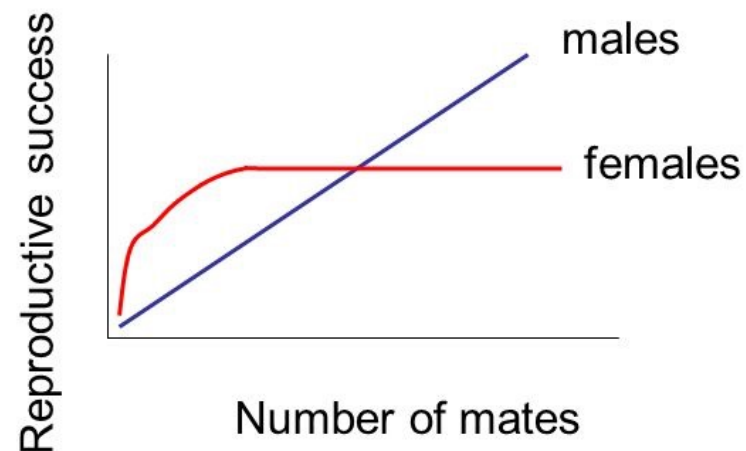




Parental investment

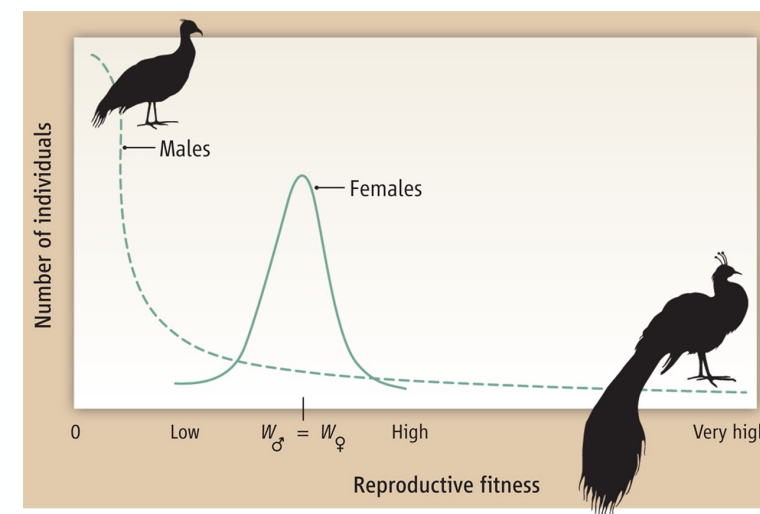


Bateman's principle (1948):



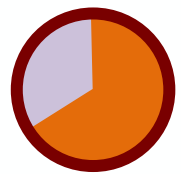
Clutton-Brock & Vincent (1991):

“the *potential* reproductive rate of males usually exceeds that of females”



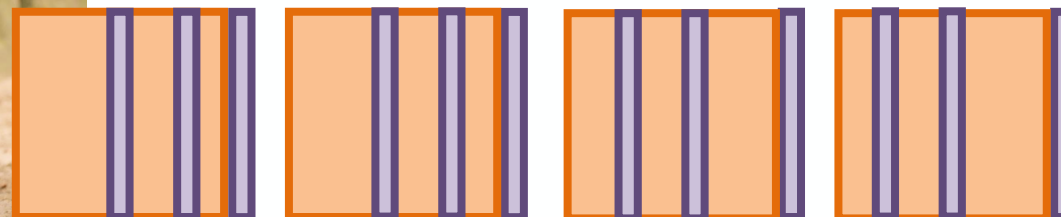
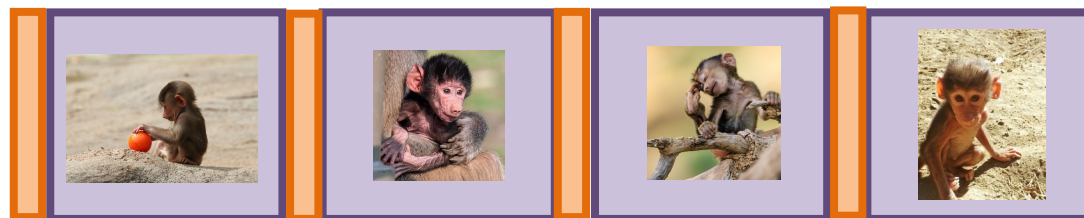
Bateman, Angus J. "Intra-sexual selection in *Drosophila*." *Heredity* 2.3 (1948): 349-368.

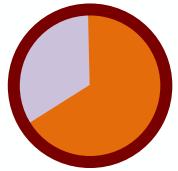
Clutton-Brock, T. H., and Amanda C.J. Vincent. "Sexual selection and the potential reproductive rates of males and females." *Nature* 351.6321 (1991): 58-60



Parental investment vs Mating effort

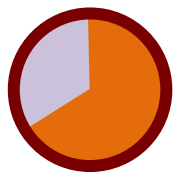
- **Parental investment** (Trivers 1972): all investment in individual offspring
- **Mating effort**: all investment in finding mates





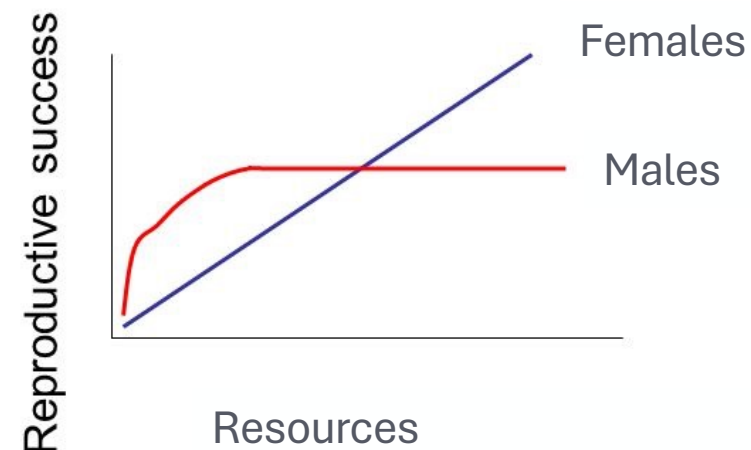
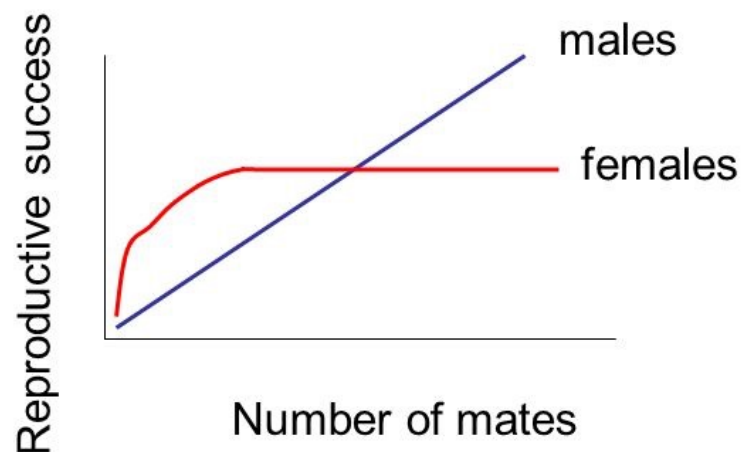
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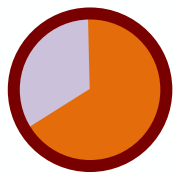
	Males	Females
Gametes	Low	High
Parental investment	Low	Often high



Parental investment vs Mating effort

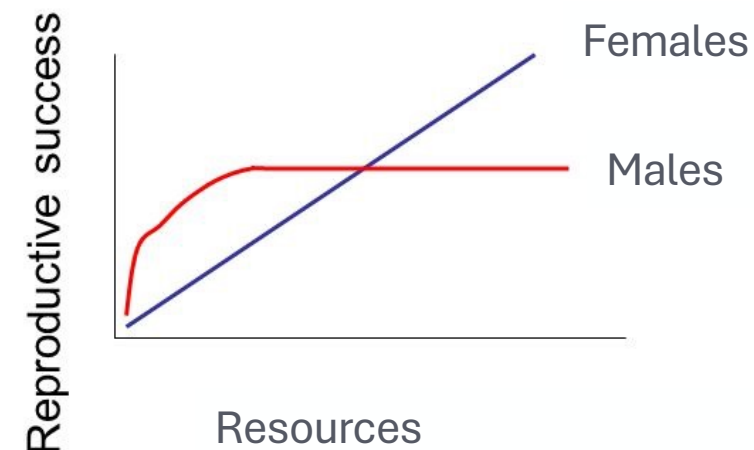
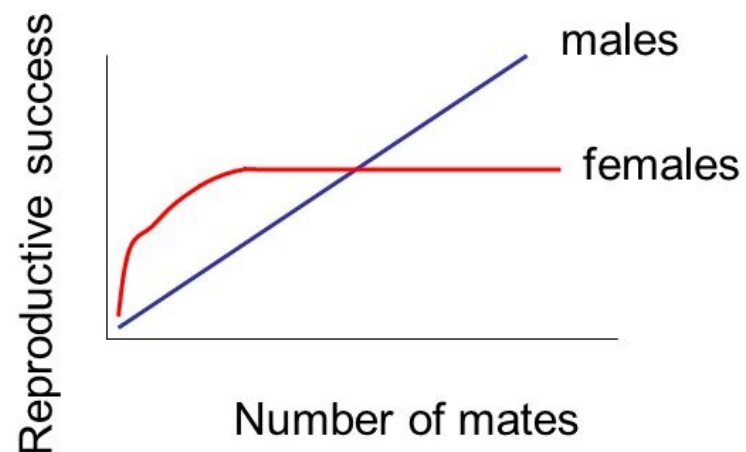
	Males	Females
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Parental investment	Low	Often high
Limiting factors	Females	Resources

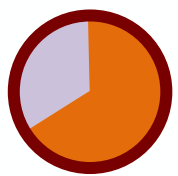




Parental investment vs Mating effort

	Males	Females
Gametes	Low	High
Parental investment	Very low	Very high
Limiting factors	Females	Resources





Parental investment vs Mating effort

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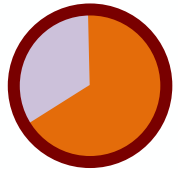
Expectations:

- Strategy focused on maximizing access to females
- Low parental investment; high mating effort
- High reproductive skew



Expectations:

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- High parental investment; low mating effort
- Low reproductive skew



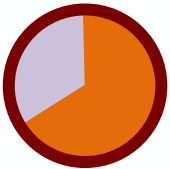
Parental investment vs Mating effort

Foetal Infancy **Childhood*** Juvenile **Adolescence***



	Males	Females
Gametes		
Parental investment		
Limiting factors		



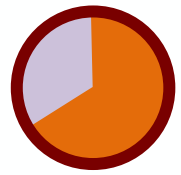


Parental investment vs Mating effort



	Males	Females
Gametes		
Parental investment		
Limiting factors		





Parental investment vs Mating effort



	Males	Females
Gametes	Low	High
Parental investment	Comparatively high	Very high
Limiting factors	Females and resources	Resources

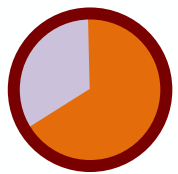


Intro

Reproductive strategies

Mate choice

Other



Parental investment vs Mating effort

Foetal Infancy **Childhood*** Juvenile **Adolescence***



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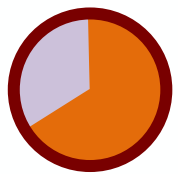
Expectations:

- Strategy focused on maximizing access to females *and* resources
- More equal balance between parental investment and mating effort
- High reproductive skew



Expectations:

- Strategy focused on maximizing access to resources
- High parental investment; low mating effort
- Low reproductive skew



Parental investment vs Mating effort

Foetal Infancy **Childhood*** Juvenile **Adolescence***



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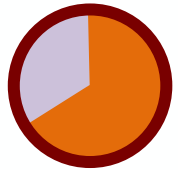
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Parental investment vs Mating effort

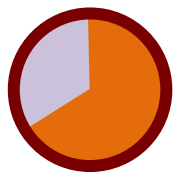
Where we fit in: Clark & Hatfield (1989)

Table 1: Study #1, 1978

Percentage of Compliance With Each Request

Sex of Requestor	Type of Request		
	Date	Apartment	Bed
Male	56%	6%	0%
Female	50%	69%	75%





Parental investment vs Mating effort

Foetal Infancy **Childhood*** Juvenile **Adolescence***



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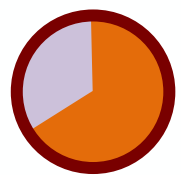
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Expectations:

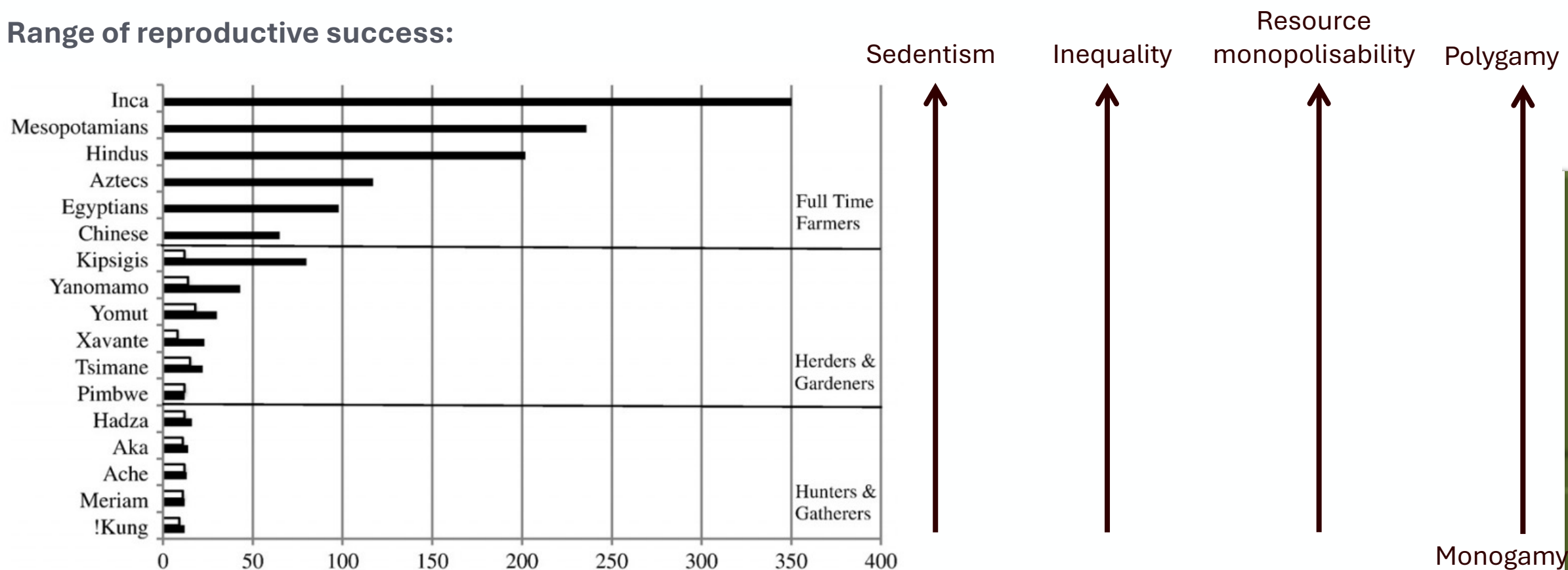
- Strategy focused on maximizing access to resources
- High parental investment; low mating effort
- **Low reproductive skew**

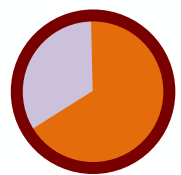


Parental investment vs Mating effort

Reproductive skew

Range of reproductive success:



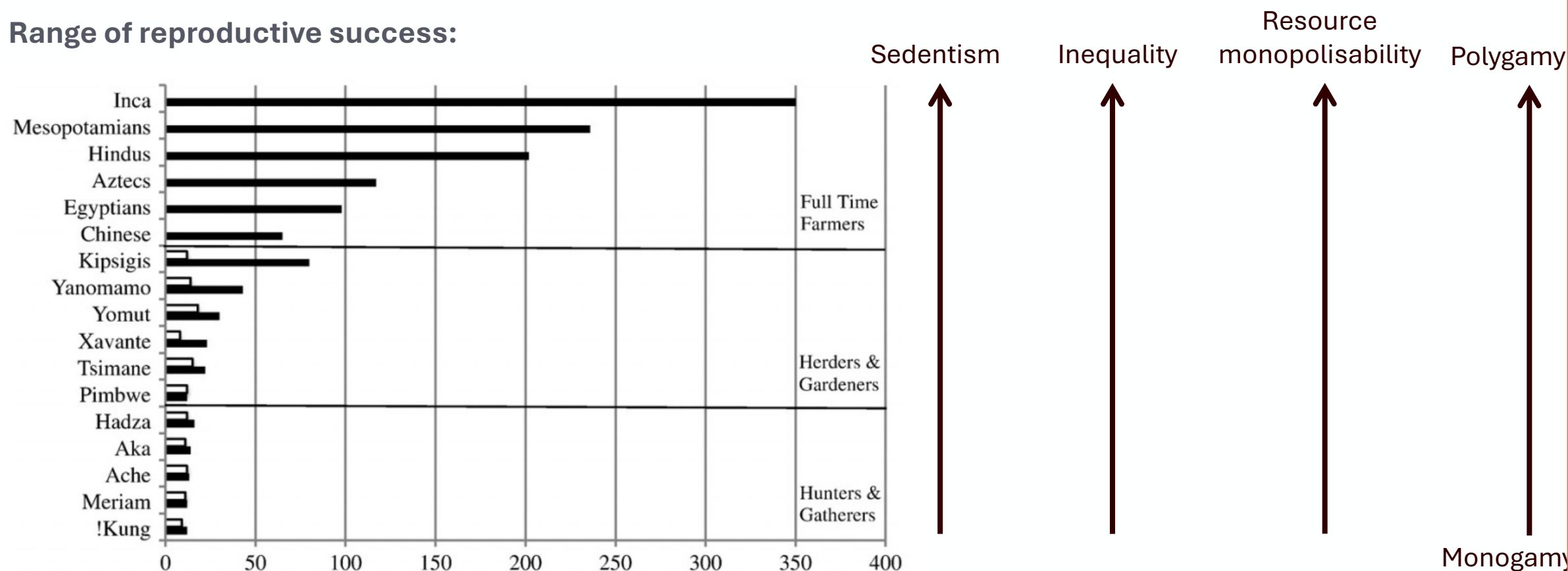


Parental investment vs Mating effort

Reproductive skew

The degree to which humans meet expectations follows social ecology

Range of reproductive success:



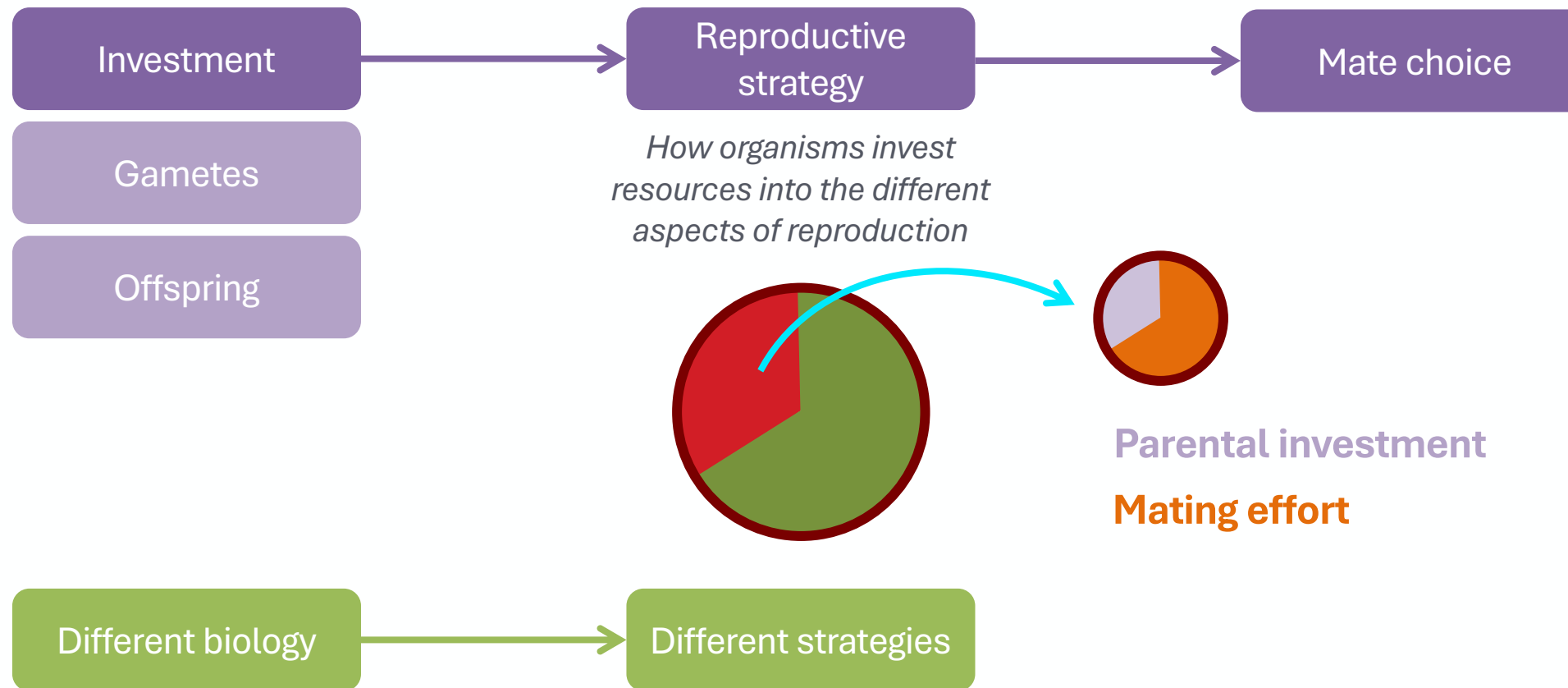
Key

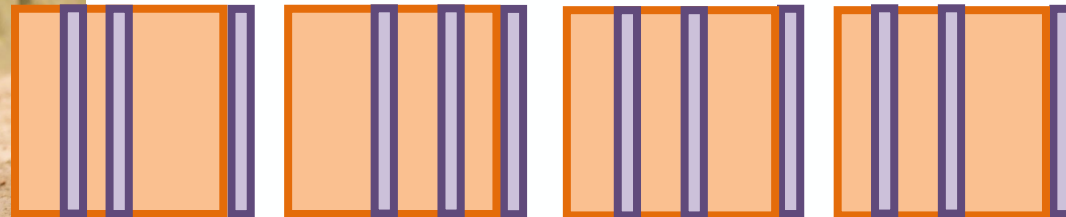
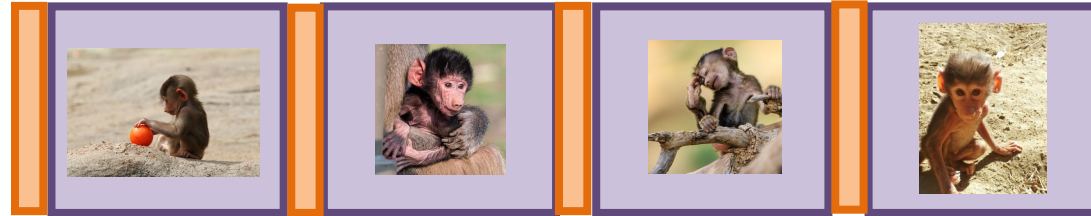
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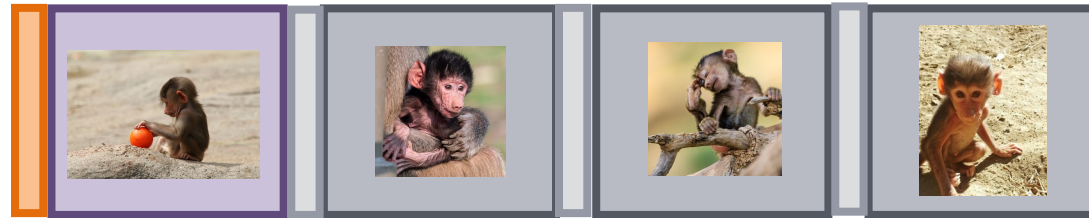
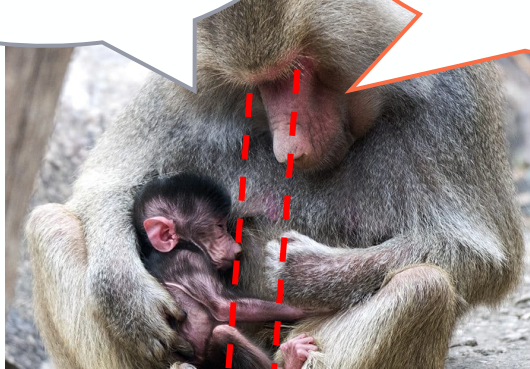




Ensure access
to resources,
and...

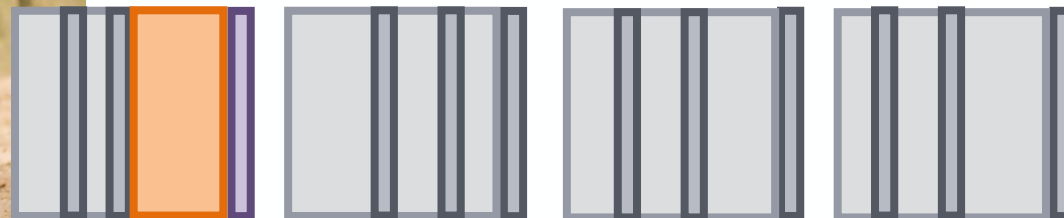
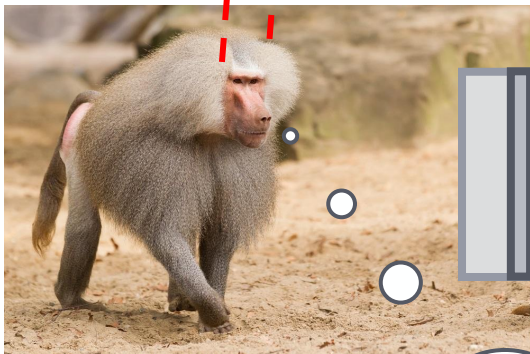
Blimey: I'd better
make sure this is a
success

Key



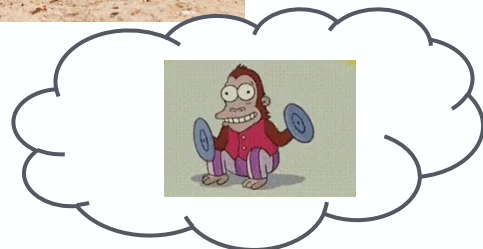
**1/4 of
lifetime
fitness**

Huge cost if
unsuccessful



**1/12 of
lifetime
fitness**

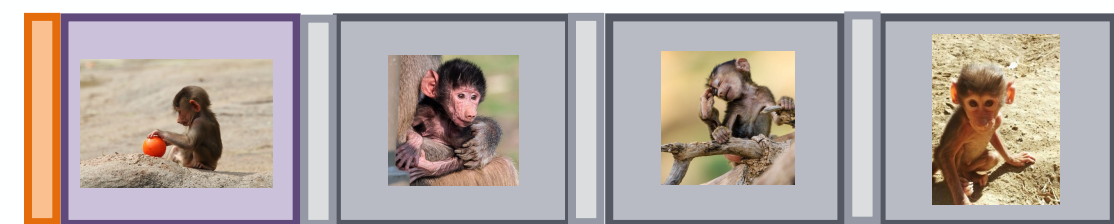
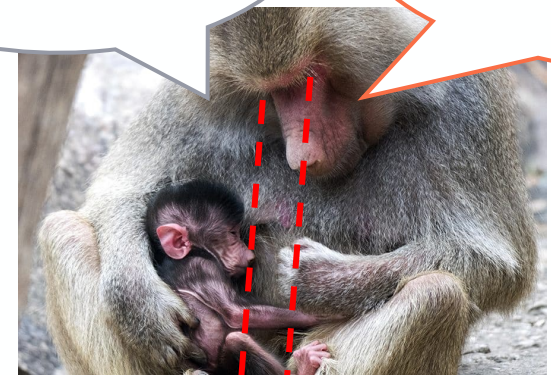
Low cost if
unsuccessful



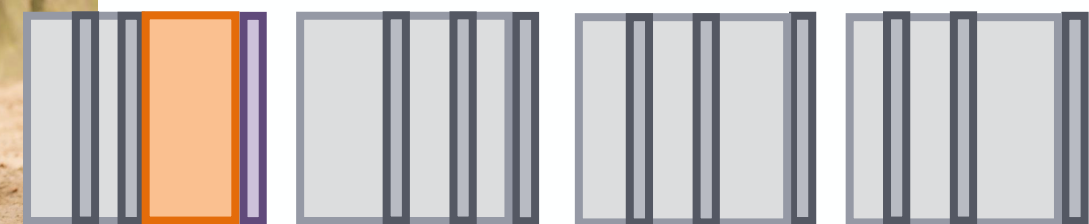
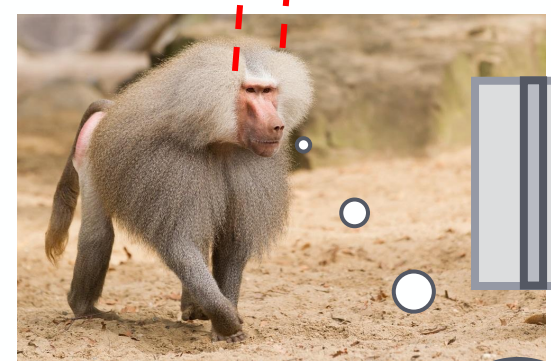
Ensure access
to resources,
and...

Blimey: I'd better
make sure this is a
success

Key



**1/4 of
lifetime
fitness**
Huge cost if
unsuccessful



**1/12 of
lifetime
fitness**
Low cost if
unsuccessful

*"Chicks for
free!"*

Because I want to reach
that high potential
reproductive success



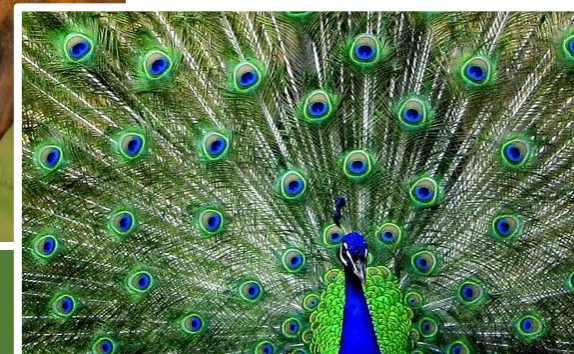
Male reproductive strategy & mate preference

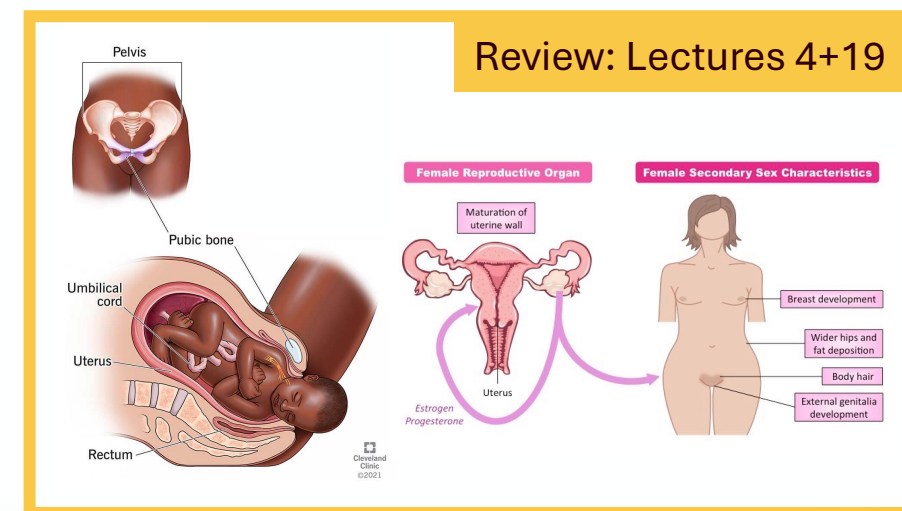
Reproductive strategies

- Low investment & high **potential** reproductive rate: high mating effort
- Males usually compete for females
- **Sexual selection: selection for traits solely concerned with increasing mating success**
 - Intrasexual
 - Intersexual

Mate preference

- **Males aren't choosy**, but may search for signals of fecundity





Adaptive Significance of Female Physical Attractiveness: Role of Waist-to-Hip Ratio

signals of fecundity

Devendra Singh

Large breasts and narrow waists indicate high reproductive potential in women

Grażyna Jasieńska^{1*}, Anna Ziomkiewicz¹, Peter T. Ellison², Susan F. Lipson² and Inger Thune^{3,4}



Female reproductive strategy & mate preference

- High investment & focused on parental investment



- **Females “choosy”**
 - Non-genetic benefits: good resources and parental ability
 - Genetic benefits

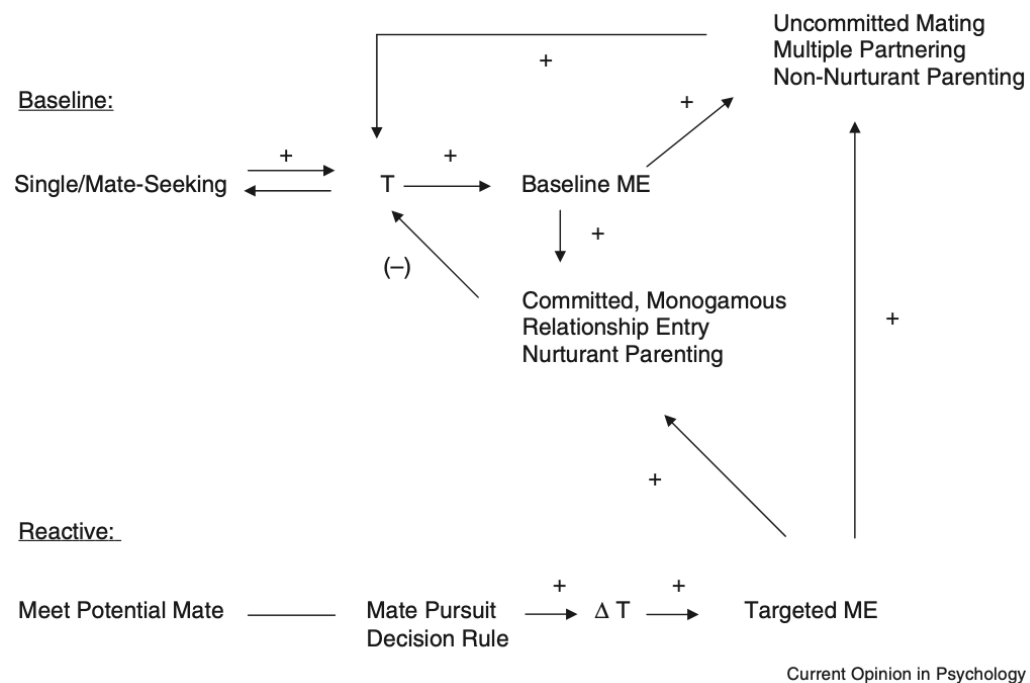




The inverse relationship between T and parental investment



The Testosterone-Relationship Cycle



High T

Low T



- Non-genetic benefits:
good resources and parental ability

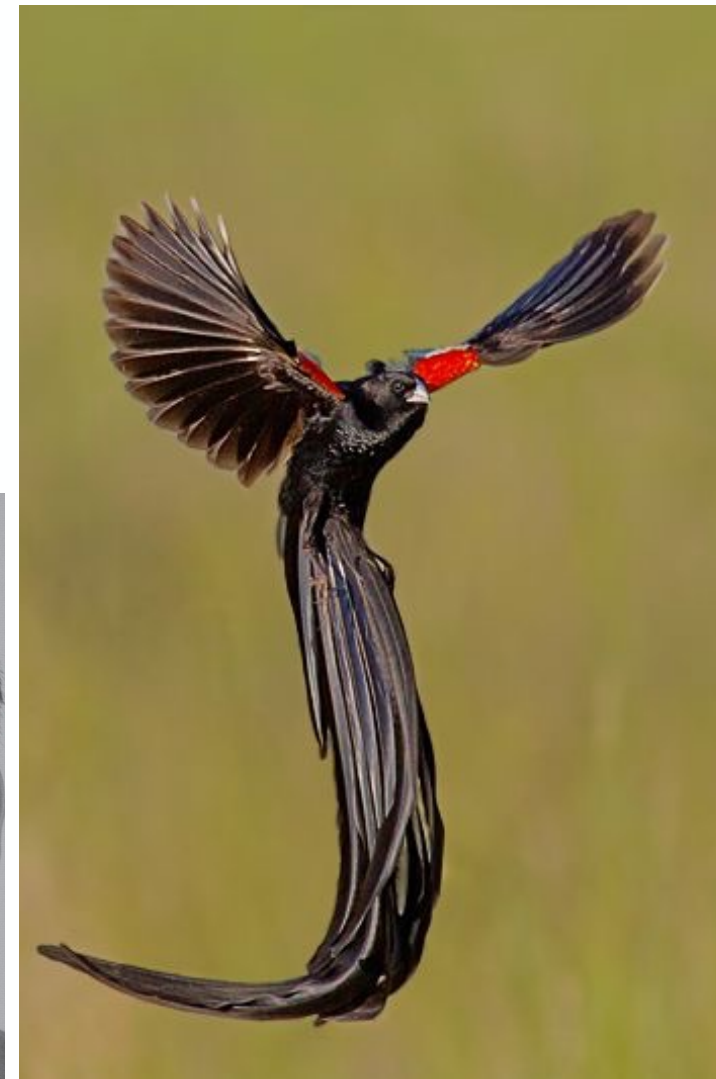


Honest signals

- **Zahavi (1975): handicap hypothesis**
- Hamilton & Zuk (1982): reliable indicators of genetic resistance to disease
- *In humans: symmetry & genetic quality*



- Genetic benefits





Expectations for mate choice preferences in humans

	Males	Females
Gametes	Low	High
Parental investment	Comparatively high	Very high
Limiting factors	Females and resources	Resources



Expectations:

- Trade-off between mating and parenting effort; balance must be sought in context of needy human babies
- Under stable reproductive bonds, males should find signals of fecundity important



Expectations:

- Needy babies: very little effort into mating, and lots into parenting
- Look for signals of
 - Willingness to invest in offspring
 - Resource access



Expectations for mate choice preferences in humans

We are about to conduct an experiment to test whether humans adhere to the patterns of mate preferences we expect based on differences in reproductive biology.

To recap, we expect males to pay attention to signals of **fertility** (e.g., waist-to-hip ratio), and we expect females to pay attention to **willingness to invest in offspring** (e.g., male access to resources, physical signals of low testosterone).

Our hypothesis is that males and females will differ in the extent to which they emphasize

1. Physical characters versus non-physical characters
2. Physical characters that convey what that sex should look for versus non-physical characters that relate to childrearing or investment in offspring

Expectations:

- Trade-off between mating and parenting effort; balance must be sought in context of needy human babies
- Under stable reproductive bonds, males should find signals of fecundity important

Expectations:

- Needy babies: very little effort into mating, and lots into parenting
- Look for signals of
 - Willingness to invest in offspring
 - Resource access



Expectations for mate choice preferences in humans

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Male sample:

	Subject 1	Subject 2	Subject 3	Subject 4	Subject 5	Total
Physical characters						
Physical characters that may convey fertility ('pretty', 'attractive', etc)						
Any non-physical characters						
Non-physical characters that relate to childrearing or investment in offspring						



Expectations for mate choice preferences in humans

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Female sample:

	Subject 1	Subject 2	Subject 3	Subject 4	Subject 5	Total
Physical characters						
Physical characters that may convey good genes or low T						
Any non-physical characters						
Non-physical characters that relate to childrearing or investment in offspring						

Mate choice preferences in humans

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Female sample:

Subject 1Subject 2Subject 3Subject 4Subject 5Total

Physical characters

Physical characters that may convey **good genes or low T**

Any non-physical characters

Non-physical characters that relate to **childrearing or investment in offspring**



Expectations for mate choice preferences in humans

When poll is active respond at [PollEv.com/lauravanholstein822](https://www.poll-ev.com/lauravanholstein822)



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Expectations for mate choice preferences in humans



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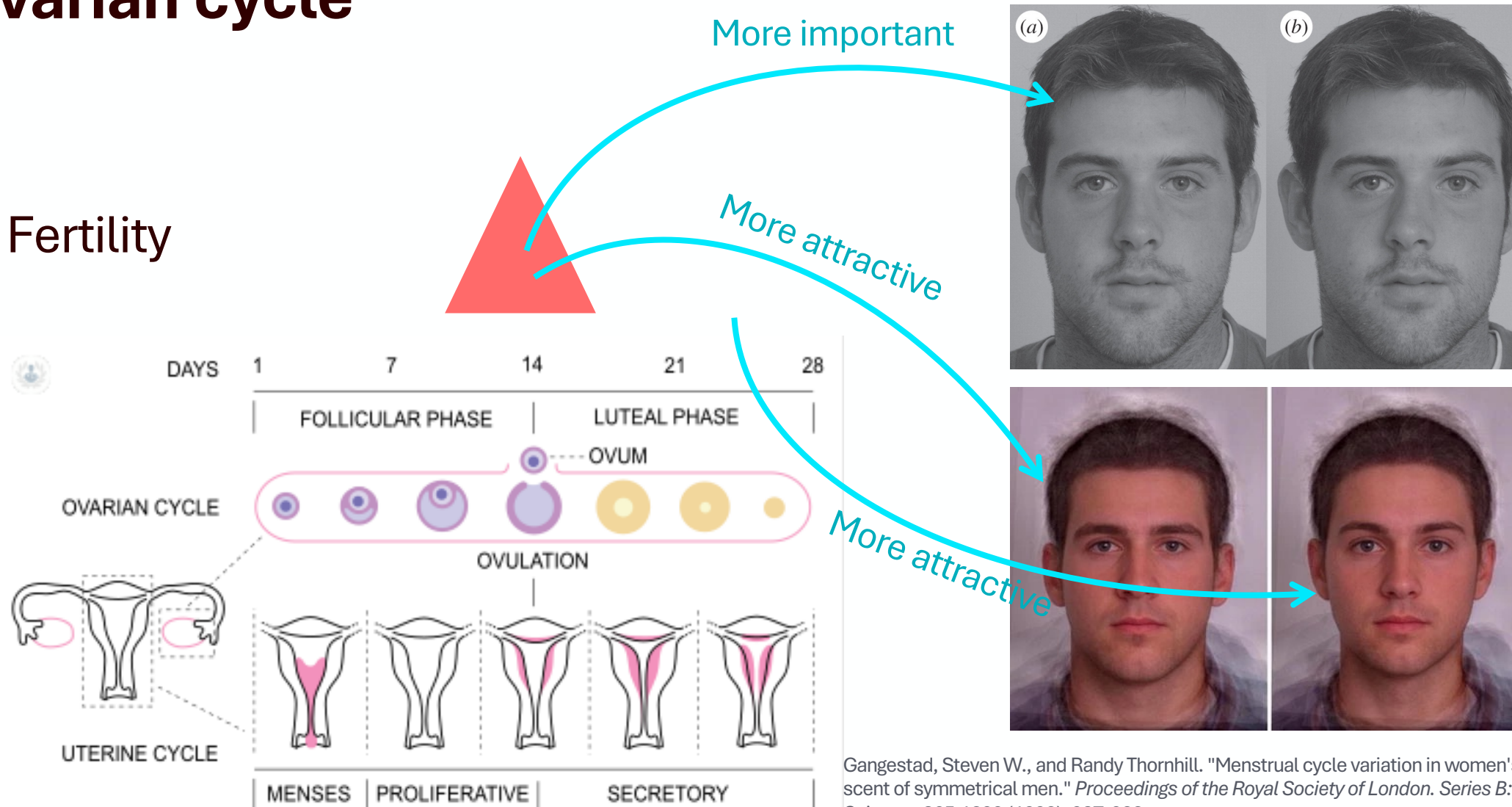
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A major confound: reproductive interests across the ovarian cycle

Fertility



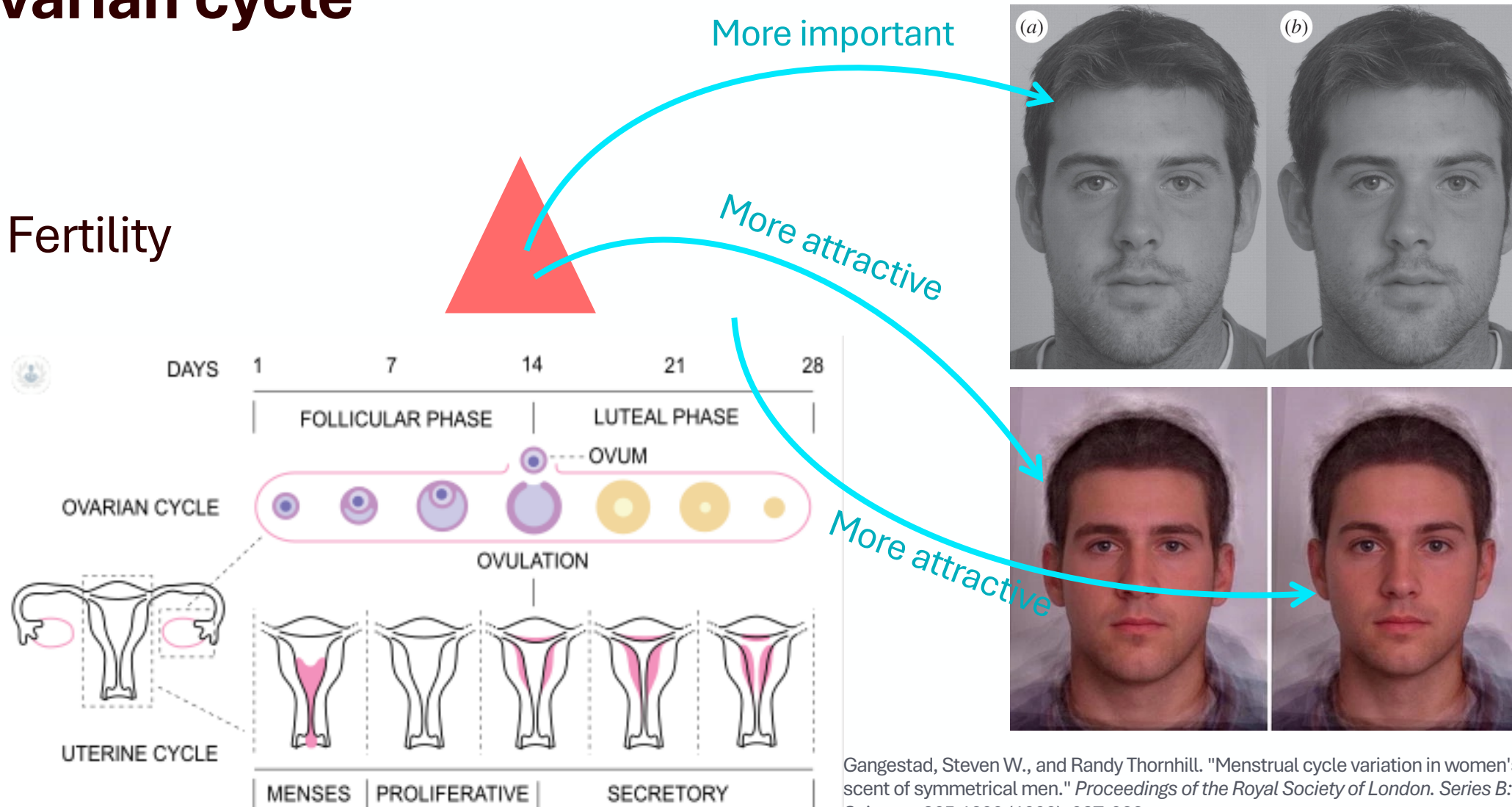
Gangestad, Steven W., and Randy Thornhill. "Menstrual cycle variation in women's preferences for the scent of symmetrical men." *Proceedings of the Royal Society of London. Series B: Biological Sciences* 265.1399 (1998): 927-933.

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A major confound: reproductive interests across the ovarian cycle

Fertility

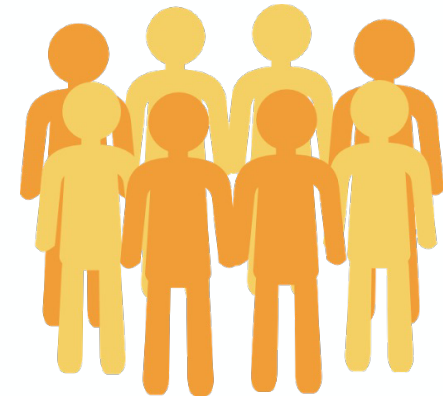
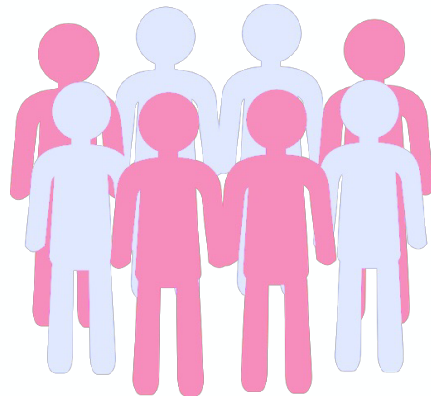
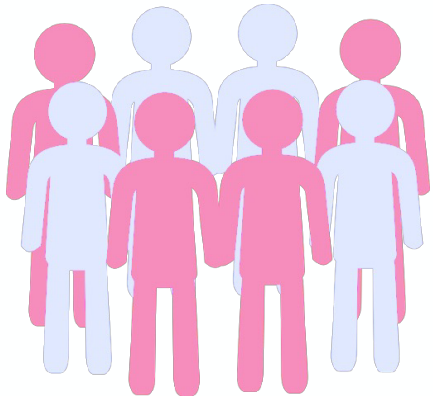


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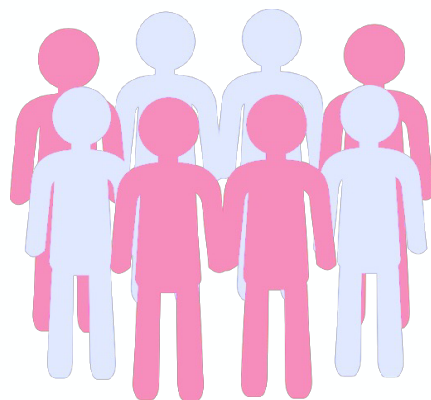
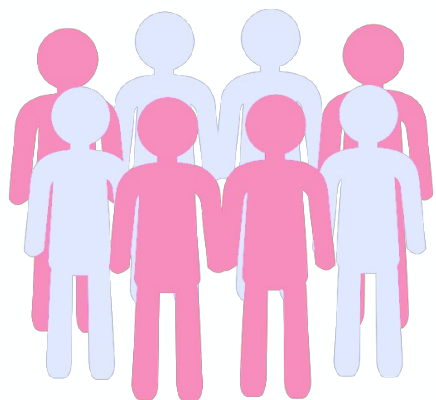
A major confound: culture





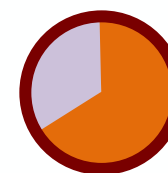
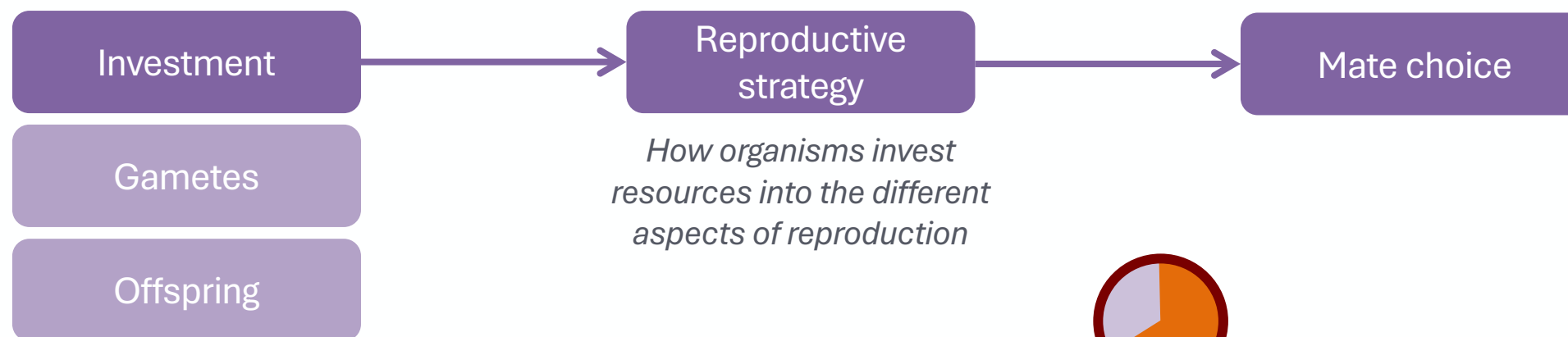
A major confound: culture

Buss (1989): Examined sex differences in mate choice in 37 populations



- In all 37 populations **men rated physical attractiveness as more important**; and in 36/37 **women rated financial prospects as more important**

- Cultural variation** found in relative importance of each variable tested



Parental investment

Mating effort



Take-home messages:

- Broadly, human averages conform to expectations, but...
 - The degree to which they do so is culturally dependent
 - Lots (and lots and lots) of variation
 - Lots of caveats



Reproductive strategy

How organisms invest resources into the different aspects of reproduction

Frequency of reproductive attempts

Investment into each offspring

Total number of offspring

Duration of pregnancy

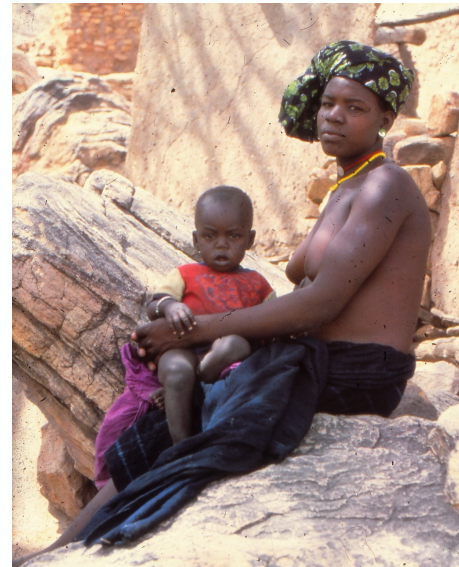
Duration of breastfeeding

Different strategies

Among the Dogon (Mali, West Africa):

Trade-off per offspring:

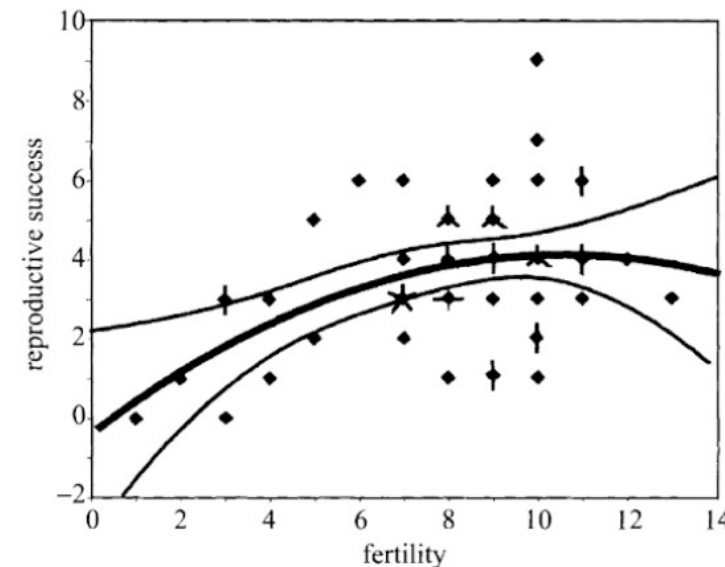
- + Reproductive success
- Chance of child mortality increases by 26% with each extra child in the nuclear unit



Total number of
offspring

Results:

- Max lifetime RS of 4.3 offspring obtained at fertility of 8.6 births.
- 75% of females achieved RS within CI of max.
- **83% of females had fertility level not significantly different from optimum**



Take-home messages:

Broadly, human *averages* conform to expectations based on our model relating reproductive traits to physiology, but...

- The degree to which they do so is **culturally dependent**
- Lots (and lots and lots) of variation within the sexes and between cultural groups
- There are many caveats. Studies often done on WEIRD populations, take humans at one moment in their lives (or menstrual cycles), and so on. Study design often doesn't allow for variability!